

GANESH P SHIVAKUMAR

Redmond, WA • ganesh2507.shivakumar@gmail.com • (617) 858-9129

[LinkedIn](#) • [LeetCode](#) • [GitHub](#) • [Portfolio](#)

Open to Relocate

SUMMARY

Experienced **Senior Machine Learning Engineer** with **6+ years** of expertise in building large-scale, high-performance distributed systems and **e-commerce** search. Proficient in **Java** and **Python**, **GoLang** with deep knowledge in **Machine Learning**, **SQL/NoSQL** databases, **BigData** processing, and cloud technologies (**AWS/Azure**). Skilled in DevOps practices. Specialize in Search & Recommendation Systems, Personalization, LLMs, and Generative AI.

Key Achievements:

- Played a pivotal role in a startup's journey from inception to its **~\$100 million acquisition**.
- Held significant positions at **Amazon**, and **Amazon Robotics**, and collaborated with dynamic teams at Microsoft.
- Led the development of ML pipelines and LLMs at **Google for Developers**.

PROFESSIONAL CERTIFICATIONS AND ACHIEVEMENTS

- Featured on the “**Employee Spotlight**” for leading an engineering team of 10 people to create tools for catalog extraction and indexing to conceptualize, design, and maintain the whole pre-processing catalog pipeline
- Received “**GoGetter & Challenger**” recognitions consistently for my persistence and problem solving ability at work
- Awarded “**Most Valuable Player**” by Unbxd, for teamwork and spearheading several teams to excel in innovation
- **Headed HackAI** (Largest AI hackathon) organized by the Artificial Intelligence Society at Northeastern University, and the third iteration of WEHack hackathon, as part of Techmakers

SKILLS

- **Agentic Coding Harnesses:** Codex, Claude Code, Kiro, Pi, Tau, DeepSeek
- **Agent Frameworks & Platforms:** Strands Agents, Amazon Bedrock AgentCore, Bedrock Knowledge Bases, SageMaker Endpoints, Model Context Protocol (MCP)
- **Multi-Agent Systems:** Agent orchestration, research and coding loops, planning, task decomposition, tool use, retries, human approval, and reusable service memory
- **Training & Evaluation:** Supervised fine-tuning, verifier and actor training, probability calibration, hard-negative training, reinforcement learning, offline policy learning, and agent evaluations
- **Inference & Research:** Recursive Language Models (RLM), vLLM, Ollama, model serving, and local and distributed inference
- **RAG & Observability:** Retrieval-augmented generation, vector search, end-to-end tracing, token usage, latency, task-completion metrics, and service-based evaluation

RESEARCH PROJECTS

Project: Verifier Bench (Training Completed | SFT | Calibration | Hard-Negative Training | Reinforcement Learning) <https://github.com/ganeshparsads/verifierbench>

- Completed verifier training using supervised fine-tuning on labeled successful and failed UI transitions, probability calibration, and hard-negative examples covering harmless visual changes, animations, and delays.
- Completed actor fine-tuning on successful Android trajectories and corrected actions, plus reinforcement learning that rewards task completion and safe recovery while penalizing unsafe actions, unnecessary verification, and false interventions.

Project: Black Hole Pi Code Editor (Recursive LM | Retrieval Policy | Offline RL | Shadow Evaluation) <https://github.com/ganeshparsads/BlackHole>

- Built an RLM trajectory logger capturing state → search/peek action → retrieved result → next action → final answer for every request, with a reward combining task success and penalties for token cost, latency, unnecessary tool calls, and incorrect or missing context.
- Established an offline-first retrieval-policy training strategy with safety evaluation, shadow deployment, and controlled online reinforcement learning; gated PPO on availability of a validated trajectory dataset and reward evaluation.

PROFESSIONAL EXPERIENCE

AMAZON GAMES (LUNAI TEAM)
Agentic AI Platform Software Engineer

Seattle, MA
Dec 2025 – Present

Project: Meta Agent (Agentic AI | Multi-Agent Orchestration | MCP | Evaluation | Observability)

- Architected a production software-engineering orchestrator coordinating eight specialized agents for task classification, deep research and critique, human-readable plan generation with Mermaid diagrams, human approval, dependency-aware task breakdown, code generation, review, and delivery through multiple GitHub pull requests.
- Instrumented every agent and tool call with end-to-end traces; published token usage, tool usage, task-completion rate, latency, failures, and retry metrics to Amazon central metrics services, with service-based evaluations and automatic recovery for failed loops.
- Built adaptive evaluation and task-classification agents that capture validated lessons as service-scoped Markdown memory for reuse; organization-wide adoption increased two-week sprint throughput from about 8 to 12 story points, a 50% improvement.

Project: Chatterbox TTS Fine-Tuning Pipeline (TTS | Model Fine-Tuning | Audio ML | AWS SageMaker | S3 | Lambda)

- Fine-tuned Chatterbox TTS models for domain-specific text-to-voice generation and built an end-to-end training, evaluation, artifact-storage, and inference pipeline using SageMaker, S3, and Lambda.

- Engineered scalable preprocessing and inference workflows that accelerated experimentation and improved deployment reliability.

Project: Knowledge Hub (Search & Discovery | AI/ML | Data Engineering)

- Architected an LLM-powered enterprise data platform supplying three Amazon GenAI services used by more than 1 million employees.
- Mentored junior engineers and drove cross-organization design reviews on distributed-systems practices.

Project: LLM Integration with Amazon Q (LLM | RAG | API Development)

- Engineered Java APIs for RAG and LLM integrations supporting more than 1 million users and 600K unique queries in three months.
- Built Java and React conversational interfaces serving 30K+ daily active users and optimized APIs to reduce latency and unnecessary call volume by 25%.

Project: Data Lake + Dashboard (ETL | Data Pipelines | AWS S3 | Analytics & BI)

- Optimized petabyte-scale telemetry ingestion with Spark and Glue ETL, built a secure S3 analytics lake, and delivered QuickSight dashboards for long-term business trend analysis.

Project: Campaign Automation Engine (FastAPI | API Gateway | Workflow Orchestration | Localization)

- Engineered FastAPI microservices to automate campaign creation, scheduling, translation, and brand review, reducing manual effort.

AMAZON WEB SERVICES (EKS TEAM)

Seattle, MA

Software Development Engineer II

Oct 2024 – Dec 2025

- Drove scalability and performance improvements on the AWS Elastic Kubernetes Service (EKS) team.
- Participated in cell-based architecture design for a large-scale distributed system that captured cluster events, transformed them, and delivered them to downstream services for billing and other business use cases. (Stack: Fluent Bit, Go, Lumberjack).
- Contributed to the Amazon EKS control plane, developing software to build, scale, and operate distributed infrastructure that helps AWS customers deploy, manage, and scale containerized applications using Kubernetes.

AMAZON ROBOTICS

Seattle, MA

Machine Learning Engineer

May 2024 – Oct 2024

- Collaborated with Applied Scientists to modify Match and Perception algorithms that map items to the correct pods in fulfillment centers.
- Designed SageMaker pipelines for continuous learning of AI models used in Match and Perception.

GOOGLE FOR DEVELOPERS (NEU STUDENT CLUB)**Boston, MA****Machine Learning Researcher****Sep 2023 – May 2024**

- Led the development of **LLMs (BERT, GPT2)**, **multi-modal variational autoencoders (VAE)** with convnets, and transformers, effectively calculating visual semantic embedding in political social media (Instagram) data with 400,000 records
- Facilitated disentangled representation learning and discovered generative factors by implementing innovative latent space exploration, partitioning it into private and shared spaces, enabling controlled robust image and text reconstruction
- Deployed the model on **GCP with advanced loss functions akin to beta-TCVAE**, achieving remarkable generative capabilities and results in generating images from text and vice versa with a 30% boost in processing, enhancing multi-modal data analysis workflows

AMAZON.COM, INC**Seattle, WA****Software Development Engineer****Jan 2023 - Sep 2023**

- Achieved 30% reduced response time by changing paginationKey serialization from orthodox string concatenation to **Kryo** on **Java-SpringBoot** based REST application
- Optimized retrieval of Quadrillion bytes of data from **Data Planes (Cairns, Andes)** by redesigning paginationKey whilst orchestrating data storage. Deployed this crucial **API** change with zero percent downtime on the existing traffic leveraging **Amazon Weblab**
- Evolutionized paginationKey which helped in modifying in-memory processing of inPredicate queries to logically evaluated queries, reducing memory usage by 50% that helped handle system overhead. (Stack: **AWS Glue, AWS Lambda, DynamoDB, Kryo**)

AMAZON ROBOTICS**Boston, MA****Software Development Engineer****Jan 2023 - May 2023**

- Pioneered a new data pipeline using **Kinesis Data Firehose - SNS - Lambda** and built a pub-sub architecture to share data snapshots among downstream services in near real time about the package details to reduce the time gap between Temporal Discrepancy Between Real-World Events and Software Execution
- Surpassed expectations by implementing in-memory snapshot solutions to reduce storage overhead, whilst achieving updated info-delivery in milliseconds through **SNS** topics
- Developed a **CDK package** for downstream services, enabling in-memory snapshots in a central storage system for package info updates, achieving 99% synchronization between the system and real-world package movement

UNBXD INC. A NETCORE COMPANY**Bengaluru, India****Machine Learning Engineer in Search****Nov 2017 - Dec 2021**

- Initiated and architected a **Kubeflow-based ML pipeline** for entity extraction (NER), personalized scoring, recommender model training/testing, and Topic Modeling, yielding a **25%** increase in customer conversion. (Stack: **Glue, Python**).
- Pioneered customer propensity scoring Learning To Rank models utilizing (ML) **Gradient Boosting Decision Trees (xgboost)** to rank products based on user analytics, which enhanced Order conversion

rates by 20% across users.

- Reduced response time by 25% by deploying an embedding-based item recommendation model trained from Factorization Machine, and serving it as an **API** on **AWS EC2** using **FastAPI** and **SCANN**.
- Implemented Attention-based **NER** models tailored for healthcare e-commerce queries, leading to a significant **50%** enhancement in search interaction and a consequent **30%** revenue increase. Additionally, built **Tableau** dashboards for visualizing **KPIs**.
- Developed Hybrid Memory-Based filtering **Recommender Systems incorporating Transfer Learning**, Graph-based, and Deep Clustering models utilizing **BERT**, resulting in **140% enhancement in MAP@5** for product recommendations.
- Coded **AutoScalable RestFul application** that validates **JSON schema**, which is extensively used in **Catalog Pipeline Streaming**. This application is built on top of **schema.org standards**. Helped in detecting irregularity in data irrespective of the size of the catalog. (Stack: **Java11, Pippo, Lombok, Maven, Docker, Kubernetes**)
- Coded **RESTful Application** in **JAVA11** for refactoring **Query Processing Layer** to centralize all the e-commerce search query rewrites and reformulations. Reduced network hops & **API latency by 20%**. (Stack: **SpringBoot, Java11, Google Guice, Maven**)
- Authored **RESTful Spring boot application in JAVA11** that acts as a source of truth for e-commerce catalogs of indexed sites, used for handling **Differential Indexing using Knowledge Graph**. Used for reducing internal overload on the search system by **20%**. (Stack: **SpringBoot, Hibernate, Java11, Neo4j**)
- Created an auto scalable platform to fetch and index catalogs from one e-commerce platform to another. An app built on **Argo** workflow where the **Kube pods** will be rolled up as per the load. (Stack: **Python, Django, Celery, PostgreSQL**)
- Coded auto scalable **Shell application** for hosting **SOLR** cloud clusters and auto recovery from various outage scenarios. This took care of **JAVA GC outages** and reduced overhead for SOLR maintenance. (**Linux, Kubernetes, Prometheus, DataDog, Grafana**)
- Mitigated data quality risks by **35%** by developing **advanced SQL** queries, **Stored Procedures**, and **Functions**. Performed ETL/ELT for direct data ingestion from diverse unit sources, and large-scale data processing using **Spark** and **MapReduce**

Software Developer

Jan 2015 - Dec 2017

- Created an auto scalable RESTful platform using **Python-GuniCorn-PostGreSql** to fetch and index catalogs from one e-commerce platform to another. Application built and deployed using **Argo workflow, Kubernetes**, and **Terraform**. Worked for multiple e-commerce platforms like Shopify, and AdobeCommerce. (Stack: **Python, Gunicorn, Celery, PostgreSQL, Swagger**)

- Designed and built **Javascript SDKs** in Builder design pattern for integrating Search, TypeAhead, Analytics, and Recommendations on e-commerce websites. This helped in reducing platform onboarding time by nearly 60%. (Stack: React JS, WebPack, Javascript (ES 6), jQuery, HandleBars)
- Developed generic **Shopify e-commerce app** in **NodeJS** with **GraphQL queries**, and **JS WebHooks** for fetching catalog from the platform. The tailored app injects CDN **ReactJS scripts** to enable custom search. Helped in reducing customer onboarding time by nearly 50 - 70%
- Leveraged **Elasticsearch**, **Logstash**, and **Kibana** to implement Analytical based Recommendation Systems for powering **personalized recommendations for e-commerce sites**. Created Kibana APM dashboards for monitoring and log analysis

EDUCATION

NORTHEASTERN UNIVERSITY

Boston, MA

M.S. Computer Science (GPA: 4.0/4.0)

Jan 2022 - May 2024

Coursework: Algorithms, Programming Design Paradigms (Java), Database Management Systems, Machine Learning, Deep Learning, Large Language Models

SRI JAYACHAMARAJENDRA COLLEGE OF ENGINEERING

Karnataka, India

Bachelors of Engineering in Computer Science (GPA: 4.0/4.0)

Jun 2011 - May 2015